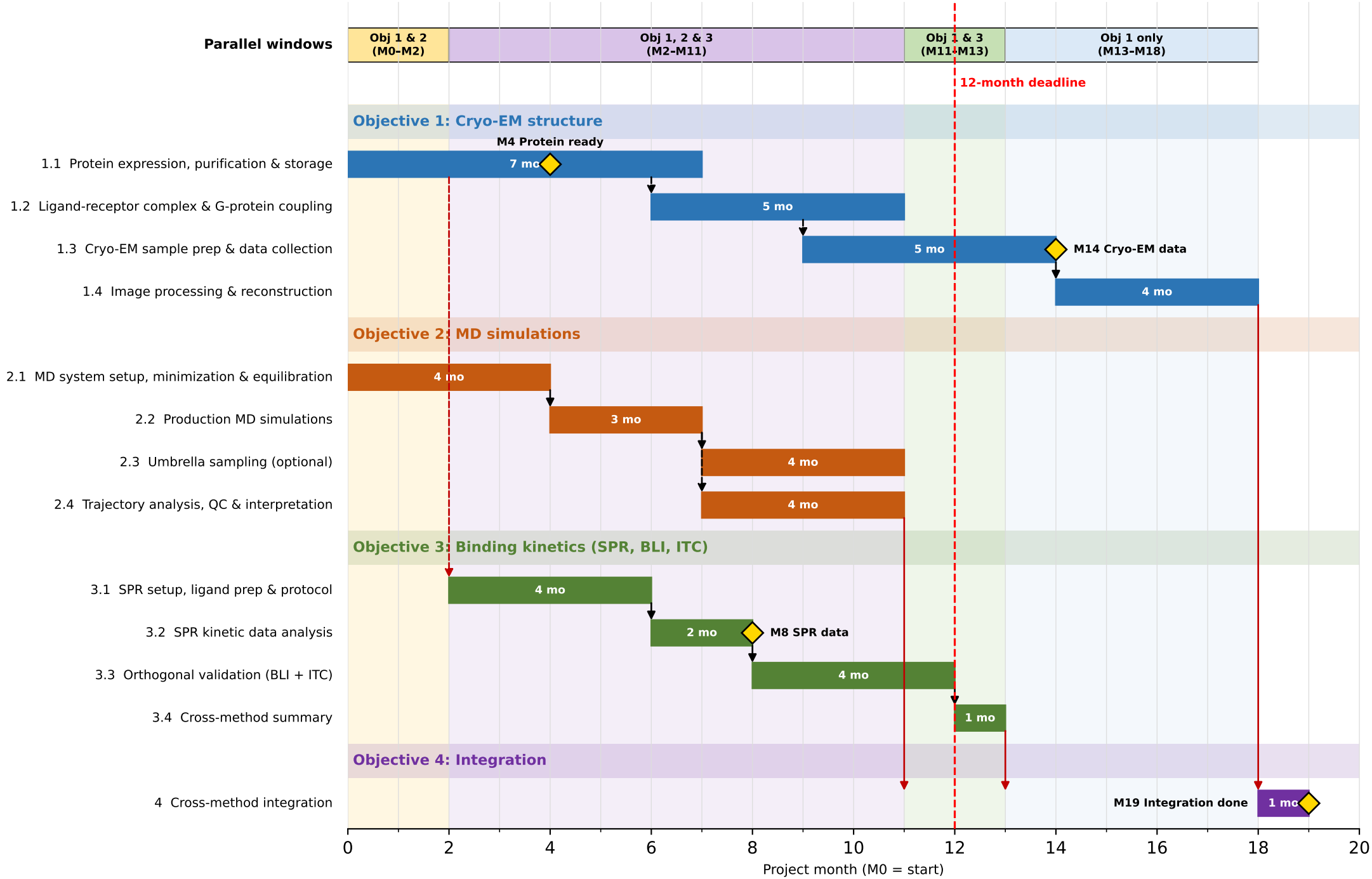
OPRM1 Ala118Gly: Objectives, Activities, Time and Connectivity

M0 is the start of the project. 'Sequence' = task waits until the predecessor ends; 'Overlap' = task starts while the predecessor is still running. The 20 original rows are clubbed into 13 activities.

Obj.	Clubbed activity	Original tasks	Activities	Time	Depends on (link type)	Dependent / Independent	Runs in parallel with (months shared)
1	**1.1** Protein expression, purification & storage	1.1.1-1.1.11	Express and purify OPRM1 (Ala118Gly), polish and store; M4 = protein ready	M0–M7 7 mo	Nothing (independent)	Independent	1.2: M6–M7; 2.1: M0–M4; 2.2: M4–M7; 3.1: M2–M6; 3.2: M6–M7
1	**1.2** Ligand-receptor complex & G-protein coupling	1.2 + 1.3	Form ligand-receptor complex and couple to G-protein; starts on first polished batches	M6–M11 5 mo	1.1 · Overlap	Dependent (same objective)	1.1: M6–M7; 1.3: M9–M11; 2.2: M6–M7; 2.3: M7–M11; 2.4: M7–M11; 3.2: M6–M8; 3.3: M8–M11
1	**1.3** Cryo-EM sample prep & data collection	1.4 + 1.5	Grid preparation and screening, then data collection; M14 = cryo-EM data	M9–M14 5 mo	1.2 · Overlap	Dependent (same objective)	1.2: M9–M11; 2.3: M9–M11; 2.4: M9–M11; 3.3: M9–M12; 3.4: M12–M13
1	**1.4** Image processing & reconstruction	1.6.1-1.6.14	Motion correction, CTF, picking, 2D/3D classification, refinement, model building	M14–M18 4 mo	1.3 · Sequence	Dependent (same objective)	
2	**2.1** MD system setup, minimization & equilibration	2.1 + 2.2	Build membrane-embedded system, minimize, equilibrate	M0–M4 4 mo	Nothing (independent)	Independent	1.1: M0–M4; 3.1: M2–M4
2	**2.2** Production MD simulations	2.3	Production runs for wild-type and Ala118Gly	M4–M7 3 mo	2.1 · Sequence	Dependent (same objective)	1.1: M4–M7; 1.2: M6–M7; 3.1: M4–M6; 3.2: M6–M7
2	**2.3** Umbrella sampling (optional)	2.4	Free-energy profiles if the production runs justify it	M7–M11 4 mo	2.2 · Sequence	Dependent (same objective)	1.2: M7–M11; 1.3: M9–M11; 2.4: M7–M11; 3.2: M7–M8; 3.3: M8–M11
2	**2.4** Trajectory analysis, QC & interpretation	2.5 + 2.6	Analyse trajectories, check convergence, interpret; starts as production ends	M7–M11 4 mo	2.2 · Overlap	Dependent (same objective)	1.2: M7–M11; 1.3: M9–M11; 2.3: M7–M11; 3.2: M7–M8; 3.3: M8–M11
3	**3.1** SPR setup, ligand prep & protocol	3.1 + 3.2	Platform setup, ligand preparation, assay protocol; runs with protein from M4	M2–M6 4 mo	1.1 · Overlap	Dependent (other objective)	1.1: M2–M6; 2.1: M2–M4; 2.2: M4–M6
3	**3.2** SPR kinetic data analysis	3.3	Fit kon, koff, KD; M8 = SPR data	M6–M8 2 mo	3.1 · Sequence	Dependent (same objective)	1.1: M6–M7; 1.2: M6–M8; 2.2: M6–M7; 2.3: M7–M8; 2.4: M7–M8
3	**3.3** Orthogonal validation (BLI + ITC)	3.4 + 3.5	BLI and ITC validation of SPR affinities	M8–M12 4 mo	3.2 · Sequence	Dependent (same objective)	1.2: M8–M11; 1.3: M9–M12; 2.3: M8–M11; 2.4: M8–M11
3	**3.4** Cross-method summary	3.6	Compare SPR, BLI and ITC results	M12–M13 1 mo	3.3 · Sequence	Dependent (same objective)	1.3: M12–M13
4	**4** Cross-method integration	4	Integrate structure, MD and kinetics	M18–M19 1 mo	1.4 · Sequence; 2.4 · Sequence; 3.4 · Sequence	Dependent (other objective)	

Longest chain: 1.1 → 1.2 → 1.3 → 1.4 → 4 (M0–M19). A slip in protein production moves cryo-EM data, the reconstruction and the final integration. Objectives 2 and 3 have slack before M18.
Milestones: M4 protein ready · M8 SPR data · M14 cryo-EM data · M19 integration complete.
Note: the chain above runs to M19, past the 12-month deadline marked on the original chart. See the assumptions flagged in the reply.

OPRM1 Ala118Gly Primary Objective: Gantt chart with dependency links (clubbed activities)

Black arrow = link within an objective; red arrow = link between objectives; solid = sequence (starts after predecessor ends); dashed = overlap (starts while predecessor runs).